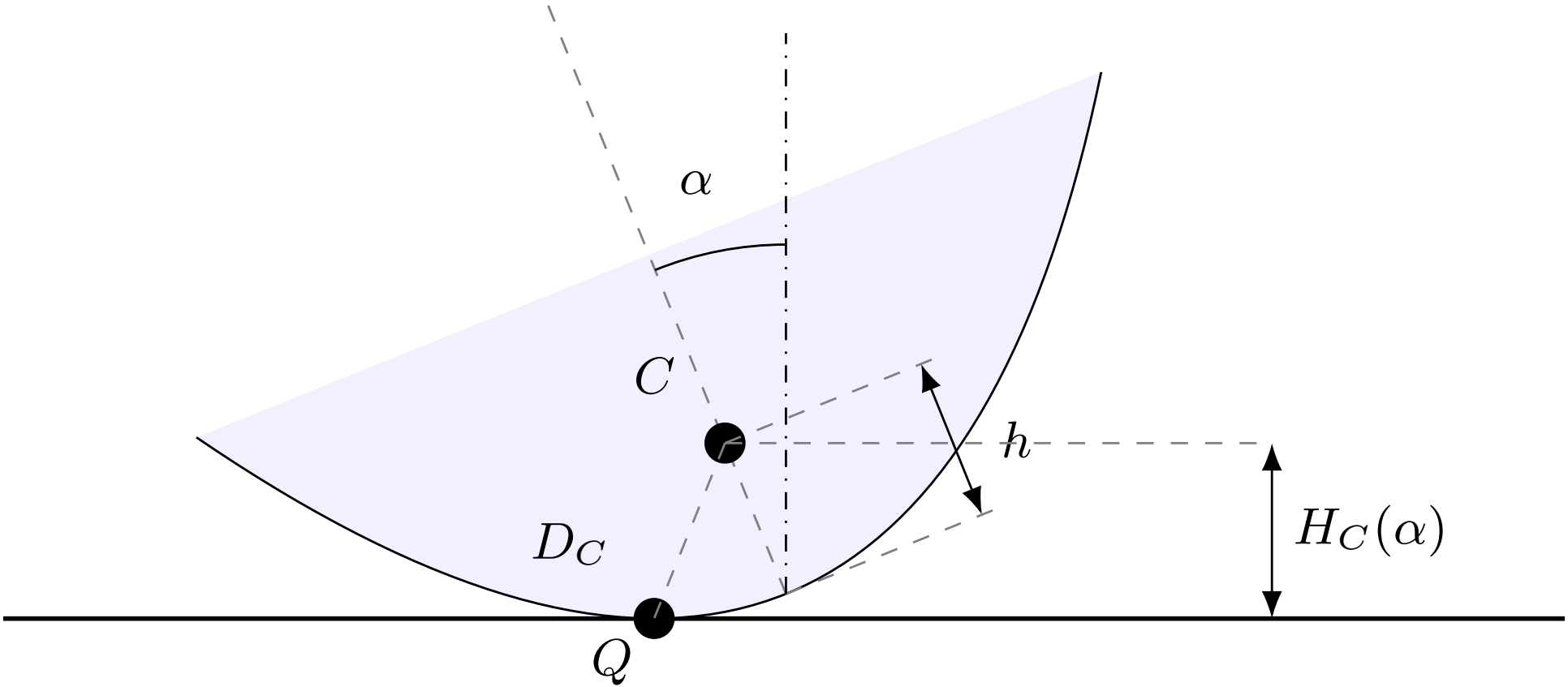

$$s_Q = -p \tan \alpha, \quad z_Q = \frac{p}{2} \tan^2 \alpha$$